

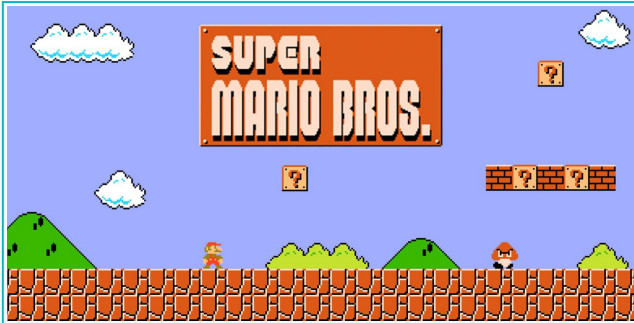


Many games involve jumping over different structures and obstacles. We control the movements of a character and move it to avoid obstacles and from falling or missing jumps. This kind of gameplay is called as platforming.

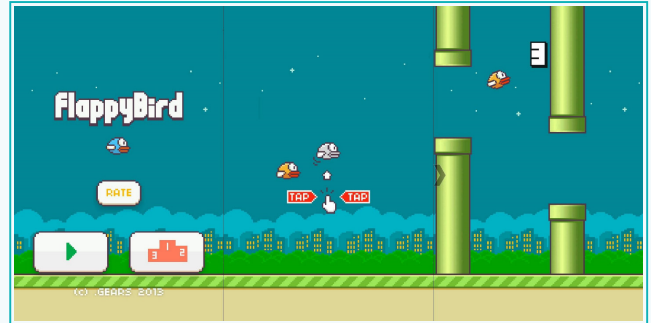
Such games are called Platformer Games.

Bytes

As long as jumping is the main part of a gameplay, any game can be called a Platformer Game.



Super Mario Bros.
Image Credit: Nintendo



Flappy Bird
Image Credit: dotGears

Platformer Games

Platformer games are the ones where the gameplay revolves heavily around players controlling a character who runs and jumps onto platforms, floors, ledges, and stairs.

Many such games include concepts like gravity and wind.

Gravity

Gravity is the force that pulls objects towards each other. Any object that has mass, also has a gravitational pull.

Earth's gravity is the reason everything stays on the ground. All objects fall to the ground because of gravity. It also determines the weight of an object, including ours.

For Example:

We stay at the ground due to gravity. When we throw objects in the air, they all eventually fall down since gravity acts upon them.



Wind

Wind is moving air. Air moves due to the difference in temperature on the Earth's surface. Uneven temperatures occur due to the Earth's rotation and its heating caused by the sun.

Wind has different speeds due to which things fly or stay in one spot.

For Example:

Wind can do beautiful things. We can watch a kite fly high in the sky due to the presence of wind.

Let's look at a real-life example where gravity and wind are both applied.



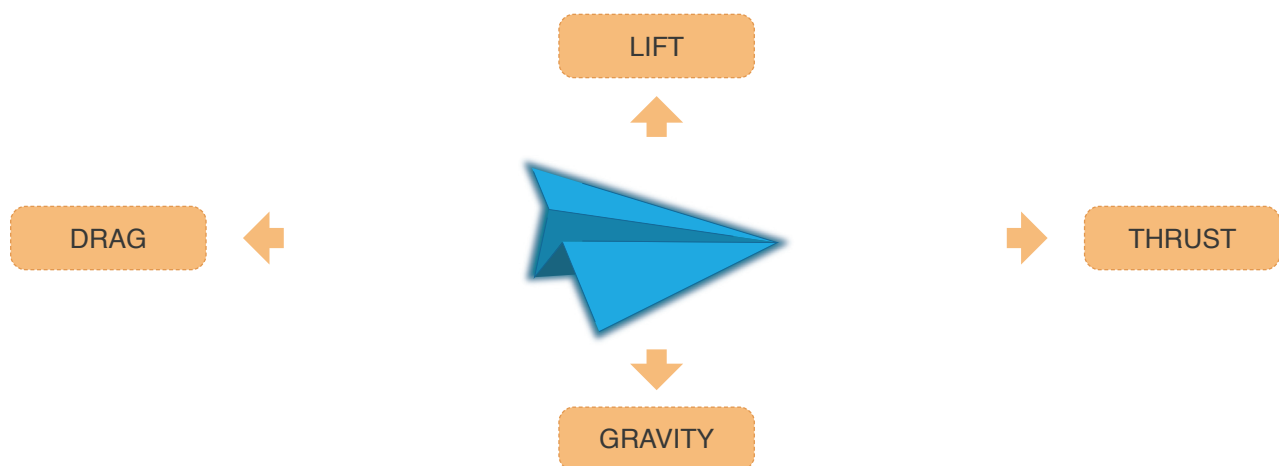
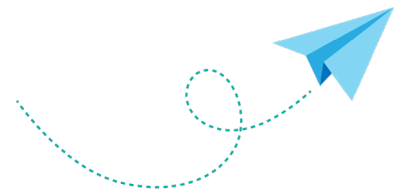
Paper planes

The air around us helps a paper plane fly. It's narrow and pointed nose cuts through the air. Air moves over and under a plane's wings to lift it up. A paper plane does not weight much, so it is easily lifted.

The weight of the plane affects its flight height and distance. This is because gravity pulls it down towards the Earth. The heavier the plane, the faster it will hit the ground.

Four basic forces act on a paper plane when it flies:

- Thrust
- Lift
- Drag
- Gravity



Game brief

In this game, the objective is to keep the paper plane flying as long as possible. It needs to be kept away from birds, trees, and other obstacles. The score will be calculated according to the distance it covers. The plane must dodge all the obstacles and the game ends if it hits an obstacle or if it touches the ground.



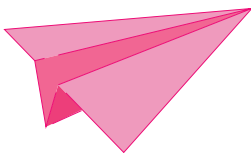
Obstacles are marked with red

Game Assets

Game assets include anything that goes into a game. This includes 2D or 3D models, sprites, and sound effects that can be used by a game engine.

These are the assets needed for our paper plane game:

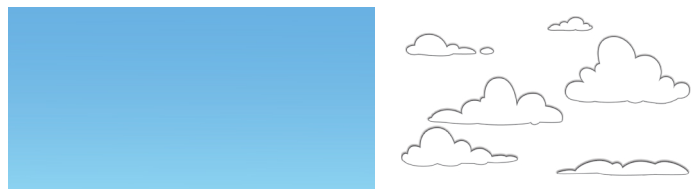
a. Player: paper plane



b. Colliders/Obstacles: ground, trees, birds



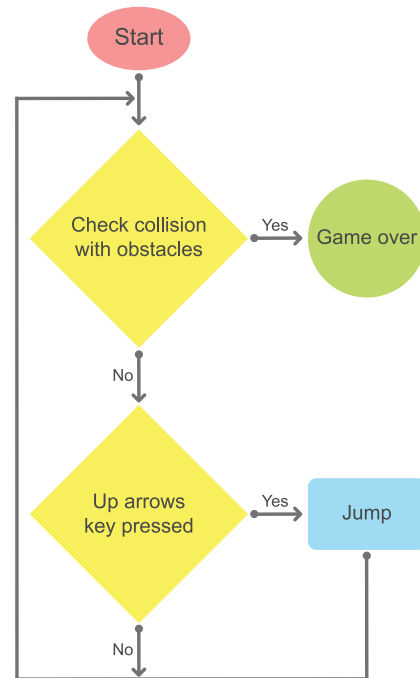
c. Backdrop: sky and clouds



Flowchart

A flowchart is a graphical representation of steps that shows the workflow to complete a task. It shows the steps involved and helps us plan the flow of the task. The steps are connected to each other with arrows that depict their order.

We will use a flowchart to create a stepwise plan for our paper plane game. This plan will help us design the game accurately.

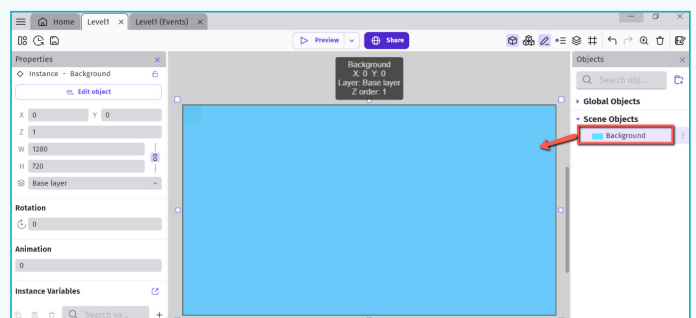
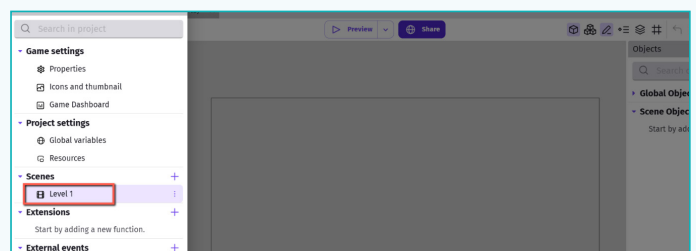


Paper plane flowchart

Activity

Use the concepts of gravity and wind to create a paper plane game.

- 1 To start, open Gdevelop. Create a new project and add a new scene.
- 2 First, let's make our own sky background to begin the game design process. To accomplish that, we must first create a sprite object, select "create with piskel," and then use the fill tool to add the blue color.

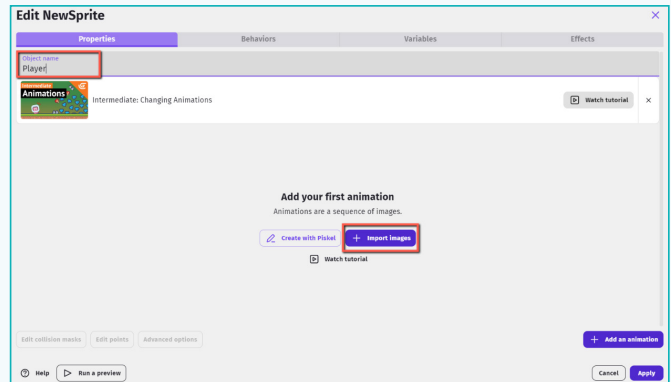


Activity

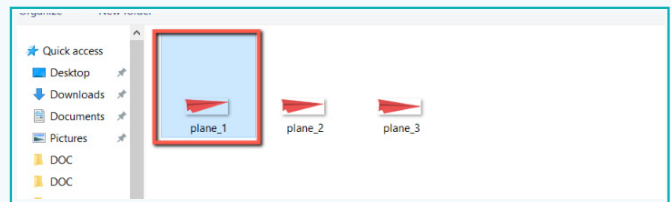
For the player character, we will use an image.

- 3 Create a new sprite object (player) of type sprite. Click on the 'import images' option.

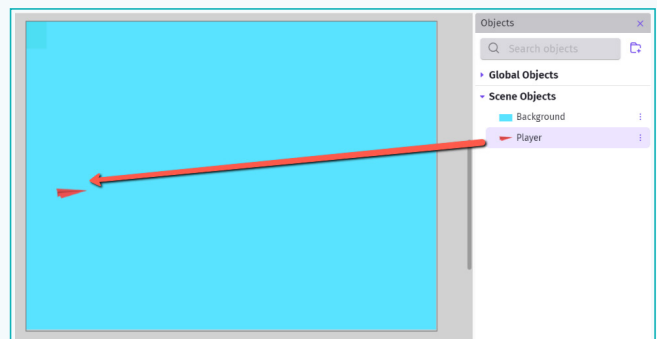
Download assets from this QR code or Link.



- 4 Select the image 'plane 1'.

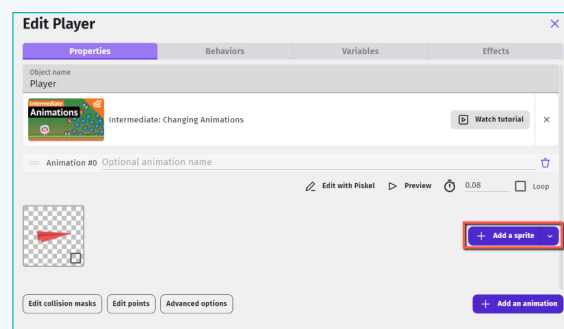


When we place the object on the scene, it will appear as a flat image.



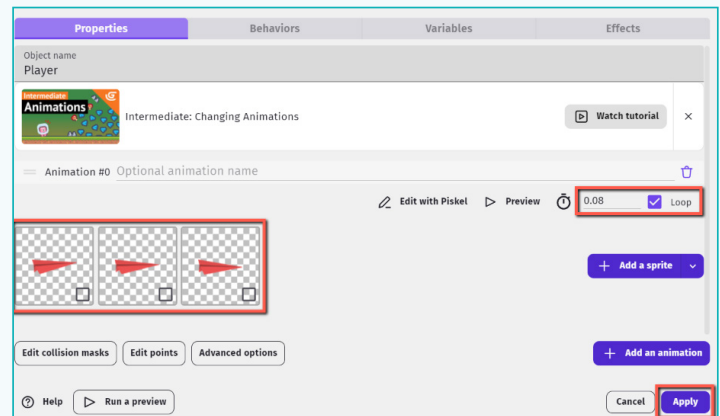
To show the plane flying in the air, we need to add some movement to it. We can use animation to create actions that emphasize this movement. We will edit the object and add multiple images.

- 5 Double click on the object to edit it. Click on 'Add' to add more images to the plane sprite.



Activity

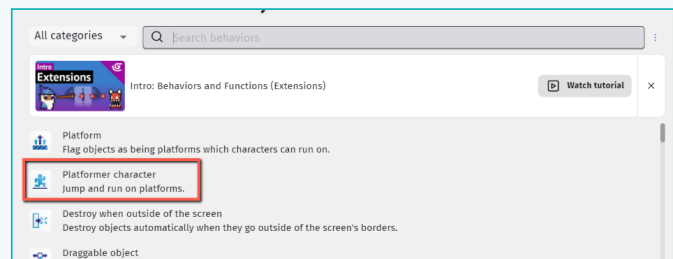
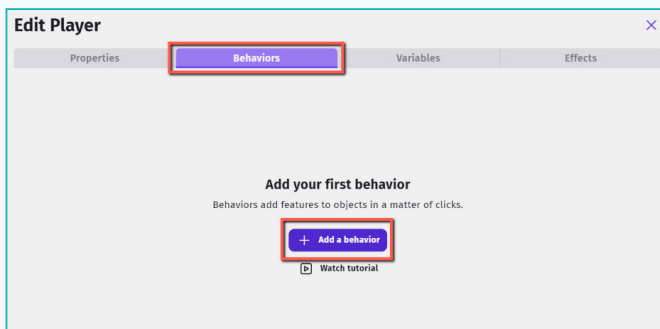
- 6 Add the other 2 plane images to the sprite object. Select the Loop option and click Apply. The loop option plays the plane images repeatedly to create an illusion of movement.



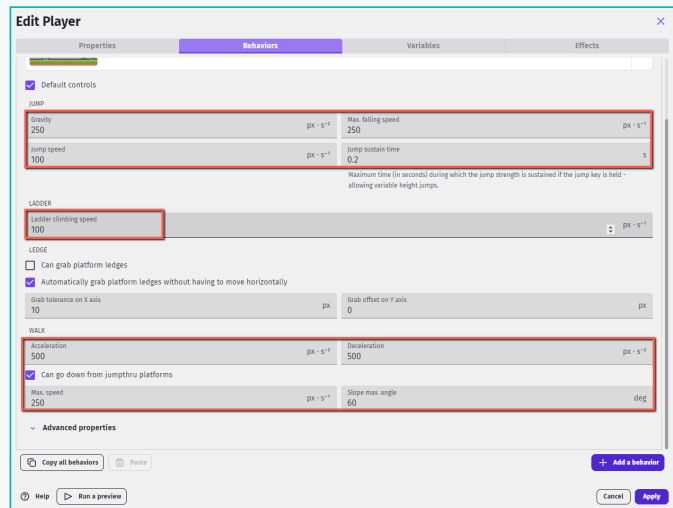
Click on the play button to test the plane's movement.

Next, we will add gravity to the player. Gravity will pull the object downward towards the ground. To achieve the gravity pull, we should use behaviors to our plane. Because these behaviors will make the objects move in a realistic way by following the laws of physics.

- 7 To add a behaviors, open the object & click on BEHAVIORS.

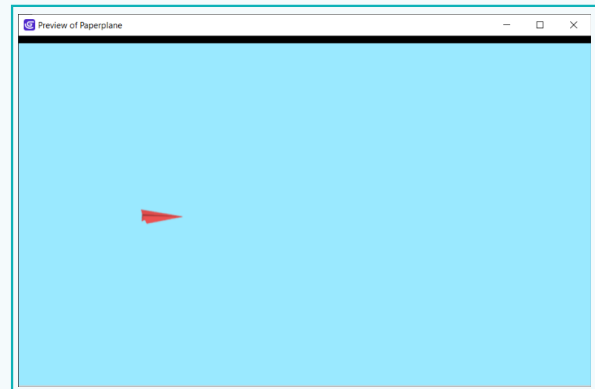


Since our character is a paper plane, the default parameters might make it difficult to play the game. We will modify the parameter strengths to suit the paper plane player.



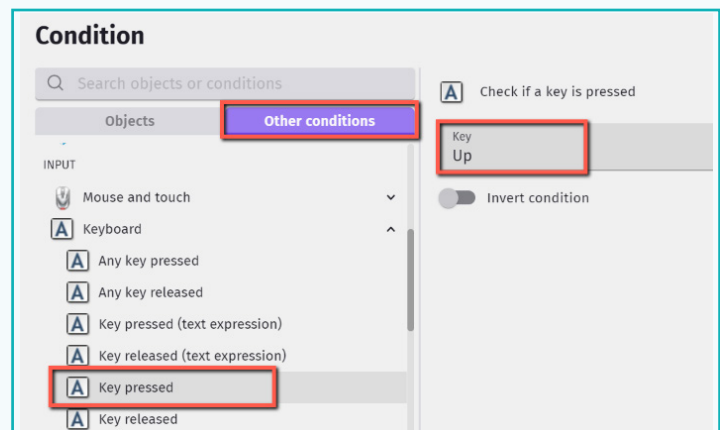
Activity

Test the behavior, the player should gradually fall down and leave the scene space.



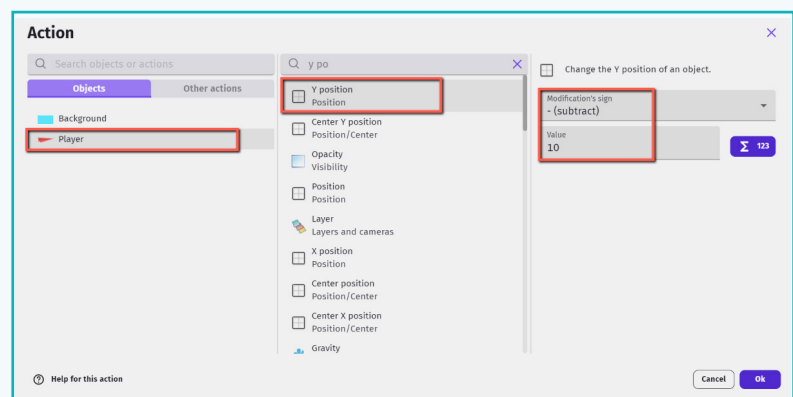
Now, let's add functionality. This will allow the player to use a keyboard to control the plane.

- 8 Add event, Keyboard- key pressed > up.



When the Up key is pressed, we want the player character to go up. We will change its position along Y axis.

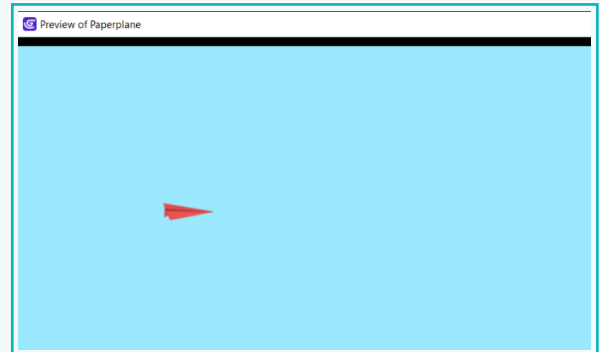
- 9 Add an action to change the vertical position(y) of a paper plane.



Activity

To see if the object is behaving the way we want, we need to test the game.

During the game's runtime, gravity should drag the player character downwards. When the up arrow key is pressed, the player will move up 10 pixels.



We can observe this mechanism in games that we play, like flappy bird. With game mechanics, we will program the plane to sustain in air as the player presses keys.

Exercise

State whether True or False

- 1 The games where we control the movements of a character and move it to avoid obstacles and from falling or missing jumps are called Platformer games.

▲ _____

- 2 Earth's rotation is the reason everything stays on the ground.

▲ _____

- 3 The force that lifts a plane up in the air is called thrust.

▲ _____

- 4 Subtracting a value for y position of an object makes the object move upwards.

▲ _____

Platformer games are those where the character stays on course until it achieves a specific target. This target can be to reach a certain score, or dodge as many obstacles as possible. Concepts of gravity and wind are implemented in such games to make them realistic.



Look sister! My highest score is 60.
Can you beat my score?

That's difficult, but I can try!

Tech Flash

- **Behaviors** : Behaviors are significant predefined features added to the objects.
- **Flowchart** : A flowchart is a graphical representation of steps that shows the workflow to complete a task.